

✓ L02 “Data is the Fuel for AI – But How Do We Use It?”

Part 1: Lab Setup & Submission Guidelines Create the notebook and name it following the guidelines: "L02_Your_Name_ITAI_2377.ipynb"

Part 2: Accessing the Dataset Clone the patitimoner/itai2377-datasets in Colab

```
!git clone https://github.com/patitimoner/itai2377-datasets.git
%cd itai2377-datasets
!ls

Cloning into 'itai2377-datasets'...
remote: Enumerating objects: 21, done.
remote: Counting objects: 100% (21/21), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 21 (delta 6), reused 16 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (21/21), 30.86 MiB | 25.45 MiB/s, done.
Resolving deltas: 100% (6/6), done.
/content/itai2377-datasets
data ITAI2377_Lab_Student_Notebook.ipynb README.md
```

Conceptual Question:

Why is it beneficial to use a shared dataset repository rather than each student downloading datasets manually?

Response: Using a shared dataset repository allows everyone accessing the data to use the same data for labwork. When you download the dataset and use that data instead of a shared dataset repository you can end up with different versions of the data, missing files, corrupted downloads, and this can change the results in the lab or make it difficult to compare work.

Part 3: Understanding Structured Data (Tabular Data)

Step 3.1: Load the Customer Dataset

```
import pandas as pd
# Load structured dataset
df = pd.read_csv("data/structured/Online Retail.csv")
# Display first few rows
df.head()
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	12/1/2010 8:26	2.55	17850.0	United Kingdom	
1	536365	71053	WHITE METAL LANTERN	6	12/1/2010 8:26	3.39	17850.0	United Kingdom	
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	12/1/2010 8:26	2.75	17850.0	United Kingdom	
3	536365	81000	KNITTED UNION FLAG HOT WATER	6	12/1/2010	3.00	17850.0	United Kingdom	

Notes on the assignment: First run returned an error for "data/structured/customer_data.csv" FileNotFoundError: No such file or directory. When I reviewed the data set, I found that there was no file with that name; however, there is an "Online Retail.csv," which is what I used for the data set.

Conceptual Question:

If an AI system were built to predict customer purchases, what kind of patterns would it need to detect?

Response:

The AI would need to detect what people buy, such as specific items, repeat items, seasonal specific items for holidays or time of year, repeat purchases, brands they favor, things purchased together, to name a few patterns.

Part 4: Understanding Image Data (Computer Vision) Unlike structured or text data, image data requires different processing.

Load CIFAR-10 Using PyTorch (TorchVision)

```
import torchvision
import torchvision.transforms as transforms
# Load CIFAR-10 dataset from TorchVision
```

```
dataset = torchvision.datasets.CIFAR10(root="./data", train=True, download=True, transform=transforms.ToTensor())
# Extract a subset of 10 images per class
class_indices = {i: [] for i in range(10)}

for idx, (_, label) in enumerate(dataset):
    if len(class_indices[label]) < 10:
        class_indices[label].append(idx)

# 🌟 TIP: What function in PyTorch DataLoader allows sampling specific indices?
```

100%|██████████| 170M/170M [00:02<00:00, 75.5MB/s]

Conceptual Question:

Why does AI need millions of images to accurately classify objects, while humans can recognize patterns with just a few examples?

Response:

AI does not process things the way humans do or can, AI detects correlations in patterns of pixels to classify objects. Humans have a lifetime of observations and experiences that let us classify objects based on these accumulated understandings of the world around us. AI models need huge amounts of data showing them multiple angles, lighting conditions, with backgrounds, without backgrounds, with distortions or partially covered just to make the correct classifications of an object.

Part 5: Understanding Text Data (Natural Language Processing)

Step 5.1: Read a Sample Text File

```
# Load and display a sample news article # 🌟 TIP: What Python function allows you to read a file?
with open("data/text/bbc-news-data.csv", "r") as f:
    text = f.read(350)

print(text)
```

"category"	filename	title	content"
"business"	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarner jumped 76% to \$1

Note on the assignment: The Hint is not exactly accurate. I did have to review the site and find the exact CSV name for the script

HINT as written: Open and analyze a sample BBC News article from GitHub (data/text/bbc_news/)

Conceptual Question:

Why do AI models struggle to understand sarcasm and context in text processing?

Response:

AI models struggle with sarcasm because sarcasm depends on the tone of your voice, the intent of what is being said, social queues and even then humans don't always grasp sarcasm. AI processes the literal meaning of the words and has learned patterns from text however sarcasm stretches the exact wording to mean something else or opposite of what is said thereby breaking the patterns.

Discussion Questions:#

How does AI process structured, image, and text data differently?

AI treats each of these data types differently because each one carries information in a different form. Structured data is stored in rows and columns with clearly defined values either numerical or in categories. This is normally provided to AI in tables and CSV spreadsheets. Images are displayed to AI as pixels in grids that it then uses convolutional networks or vision transformers to understand and find the patterns in to identify the images. Text data is unstructured and comes from books, newspapers, and various forms of written works. AI stores this data as tokens and learns the relationships between the words to make predictions.

What real-world AI applications combine multiple types of data (structured, image, text)?

Healthcare Diagnostics is a great example of real-world AI applications combining structured, image, and text. Images like X-rays and MRIs get processed with structured patient data like numerical values taken during visits where height, weight, blood pressure and the like all get documented. This is also processed with doctors notes, medical history, and medical books to help diagnose or suggest treatment options.

Imagine you're building an AI personal assistant. What type of data would it need?

My AI assistant would need access multiple forms of data to help me get to where I need to be, when I need to be and make predictions on what I may need in the immediate future. This would involve access to my calendar events and reminders, contact lists, messages, emails, current location tracking, and local time tracking. These types of data are structured, unstructured and dynamic.